Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/13/2005
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Project Title:	Two-Fluid Model and Interfacial Area Trasport in Microgravity Condition		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Sun, Xiaodong (The Ohio State University) Mi, Ye (En'Urga Inc.)		
Grant/Contract No.:	NNC04GA26G		
Performance Goal No.:			
Performance Goal Text:			
	Two-fluid model is considered as the most detailed model to describe two-phase flow problems. An accurate constitutive relation for the interfacial area concentration is essential for the two-fluid model to achieve high accuracy. Traditionally, the interfacial area concentration has been specified based on flow regimes and the transition criteria, which do not dynamically represent the changes in the interfacial structure. The flow regime maps are based on the assumptions of steady-state and fully-developed flows. However, under microgravity condition, two-phase flow may not reach a steady-state condition and the interfacial structure does not reach an equilibrium configuration, particularly at low flow condition. Therefore, the use of flow regime dependent closure relations developed for the fully-developed steady-state two-phase flow may cause significant errors, in particular for microgravity two-phase flows. In view of this, a completely novel approach that takes into account the dynamic two-phase flow structure by		

Task Description:	introducing interfacial area transport equation into the two-fluid model is being studied. The interfacial area transport equation has the potential for improving the predictions as well as eliminating the multiple bifurcation phenomena. The interfacial area transport equation is formulated by mechanistically modeling the physical processes that govern the creation and destruction of interfacial area. Therefore, it dynamically models the changes in the flow structure. Recently, the interfacial area transport equation has been successfully developed for the normal gravity condition (Wu et al., 1998; Kim, 1999; Sun, 2001, Ishii et al., 2002). It was demonstrated that by using the interfacial area transport equation, the interfacial structures could be dynamically predicted along the flow field. Furthermore, the preliminary code implementation study showed a leapfrog improvement in the code prediction (Ishii et al., 2001). The objective of the present research is to develop the two-fluid model with the interfacial area concentration transport equation applicable to the microgravity condition. In the proposed study, the two-fluid model for the microgravity two-phase flow condition will be formulated. In addition, the interfacial area transport equation for the microgravity condition will be established through mechanistic modeling of dominant fluid particle disintegration and coalescence phenomena. Along with the theoretical studies, a detailed experimental study is also proposed to perform a fundamental investigation on two-phase flow structure under microgravity condition and to establish a database. As the first phase of the experimental investigations, a ground-based experiment with two fluids of similar densities will be employed to simulate the microgravity condition. The detailed two-phase flow parameters will be acquired by the state-of-the-art global and local two-phase flow measurement techniques, such as advanced image processing, multi-sensor impedance void-meter, and local four-sensor conductivity probe. The database established in the experiment will be used to benchmark the theoretical models. The objective of this study is to provide a better predictive tool for microgravity gas-liquid two-phase flow. The expected final products of the proposed study are: 1. Development of framework of the two-fluid model formulation with interfacial area transport equation applicable for microgravity environments; 2 . Development of constitutive relations for interfacial area transport equation under microgravity condition through mechanistic modeling of fluid particle interactions; 3. Scaling study and construction of simulation experiments for the interfacial area transport experiment for microgravity two-phase flow; 4. Database of detailed two-phase flow parameters from the state-of-the-art local and global measurements; and 6. Evaluation of the interfacial area transport equation for microgravity condition based on the acquired database. Mechanistic modeling of particle interaction mechansims and therefore flow structure development and Dynamic predictive capability for two-phase flow in microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Better understanding of particle interaction mechanisms which are difficult to isolate otherwise; Extension of the interfacial area transport equation and two-fluid model to predict different class of flows; Extensive technical expertise and database for earth based simulated microgravity experiments.
Task Progress:	- Framework of the technical approach and mechansitic modeling has been identified; - Experimental facility with 25 mm inner diameter (ID) test section has been constructed; - Scoping/Exploratory tests has been carried out in 25 mm ID test section; - Experiments are being performed to collect local interfacial structure data; - Possible fluid particle interaction mechanisms have been identified; - A larger size test section (300 mm ID) is being designed and construction will be completed by the end of the current fiscal year; - Experiments will be performed in the larger test section to study the bubble size limits (spherical, cap, and maximum stable) as well as local data of interfacial structures; - Surface tension effect will be accounted in the two-fluid model for microgravity applications.
Bibliography Type:	Description: (Last Updated: 07/20/2007)
Abstracts for Journals and Proceedings	Vasavada, S., Sun, X., Ishii, M. and Duval, W., "Investigation of Two-phase Flows in Microgravity using Ground Based Experiments," 20th Microgravity Science and Space Processing Symposium (Embedded to the 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 9-12, 2006). Microgravity Two-phase Flow, Interfacial Structures, Interfacial Area Concentration, Interfacial Area Transport Equation , Jan-2006
NASA Technical Documents	Vasavada, S.; Sun, X. and Ishii, M., "A Technical Progress Report on the Study of Two-fluid Model and Interfacial Area Transport Equation in Microgravity Condiiton," NASA Technical Report. Sep-2005
Presentation	Vasavada, S.; Sun, X.; Ishii, M. and Duval, W., "Interfacial Structures of Two-phase Flow in Simulated Microgravity Conditions," Space Nuclear Conference 2005, Embedded to the American Nuclear Society 2005 Annual Meeting, San Diego, CA, June 6-9, 2005. Jun-2005